

Reg. No.

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

BCACACS 303

**Third Semester B.C.A. Degree Examination, December 2025/January 2026
(SEP) (2025-2026 Batch Onwards)
COMPUTER NETWORKS**

Time : 3 Hours

Max. Marks : 80

Note : Answer **any ten** questions from Part – A and **one full** question from **each** Unit of Part – B.

PART – A

Answer **any ten** questions.

(10×2=20)

1. a) Define computer network.
- b) Expand TCP and UDP.
- c) What is unicasting and broadcasting ?
- d) What is meant by unguided media ?
- e) What is path loss ?
- f) What is multiplexing ?
- g) What is routing algorithm ?
- h) Write the various services offered by the network layer to the transport layer.
- i) Expand PPP and HDLC.
- j) List the two main protocols in transport layer.
- k) What is name resolution ?
- l) What is WebMail ?



P.T.O.

**PART – B****Unit – I**

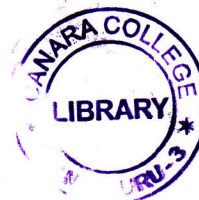
2. a) Explain in brief different layers of OSI reference model.
b) Write a note on the following :
 - i) Business application
 - ii) Home application.
- c) Differentiate connection oriented and connectionless service. **(5+5+5)**
3. a) Explain in brief different layers of TCP/IP reference model.
b) Explain design issues for the layers.
c) Write a note on service primitives. **(5+5+5)**

Unit – II

4. a) Write a note on twisted-pair cable.
b) Write a note on the following :
 - i) Radio transmission
 - ii) Light transmission.
- c) Explain Hamming code of error-correcting codes. **(5+5+5)**
5. a) Write a note on the following :
 - i) Coaxial cable
 - ii) Fiber optics
- b) Explain Checksums of error-detecting code with an example.
c) Explain the different layers of switching techniques. **(5+5+5)**

Unit – III

6. a) Explain store and forward packet switching.
b) Explain the concept of Flooding.
c) List and explain the principles of network layer in the internet. **(5+5+5)**
7. a) Explain hierarchical routing with an example.
b) With a neat diagram, explain the format of IPv4 protocol.
c) Explain classful addressing. **(5+5+5)**





Unit – IV

8. a) Explain the process of connection establishment in transport layer.
b) Explain the architecture of Email system.
c) Write a note on SMTP. (5+5+5)
9. a) With a neat diagram, explain user datagram protocol.
b) Explain TCP service model.
c) Explain the format of internet message. (5+5+5)

